

Spice Model

5.4

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Module Index

1.1 Modules

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Chapter 2

Namespace Index

2.1 Namespace List

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3.1 Class Hierarchy

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Data Structure Index

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File Index

5.1 File List

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Chapter 6

Module Documentation

6.1 Models

Modules

- [Environment](#)

6.1.1 Detailed Description

6.2 Environment

Modules

- [Spice](#)

6.2.1 Detailed Description

6.3 Spice

Files

- file [spice_ephem.hh](#)
Define class for the SPICE ephemeris model.
- file [spice_ephem_orient.hh](#)
Define class SpiceEphemOrient, which extends EphemerisOrientation for use with the SPICE ephemeris model.
- file [spice_ephem_point.hh](#)
Define class SpiceEphemPoint, which extends EphemerisPoint for use with the SPICE ephemeris model.
- file [spice_ephem.cc](#)
Define the methods for the SPICE ephemeris model class.
- file [spice_ephem_orient.cc](#)
Define the methods for the SPICE-specific ephemeris orientation class.
- file [spice_ephem_point.cc](#)
Define the methods for the SPICE-specific ephemeris point class.

Variables

- static constexpr int `MAX_PATH_LENGTH` = 129
- static constexpr int `MAX_NAME_LENGTH` = 33
- static constexpr int `MAX_MSG_LENGTH` = 1841
- static constexpr int `MAX_IDS` = 1000

6.3.1 Detailed Description

6.3.2 Variable Documentation

6.3.2.1 MAX_IDS

```
constexpr int MAX_IDS = 1000 [static], [constexpr]
```

Definition at line 55 of file `spice_ephem.cc`.

Referenced by `jeod::SpiceEphemeris::process_spk()`.

6.3.2.2 MAX_MSG_LENGTH

```
constexpr int MAX_MSG_LENGTH = 1841 [static], [constexpr]
```

Definition at line 54 of file `spice_ephem.cc`.

Referenced by `jeod::SpiceEphemeris::load_spice_files()`, and `jeod::SpiceEphemeris::update_trans()`.

6.3.2.3 MAX_NAME_LENGTH

```
constexpr int MAX_NAME_LENGTH = 33 [static], [constexpr]
```

Definition at line 53 of file `spice_ephem.cc`.

Referenced by `jeod::SpiceEphemeris::add_barycenter()`, `jeod::SpiceEphemeris::mute_spice_errors()`, `jeod::↔
SpiceEphemeris::name_barycenter_frames()`, and `jeod::SpiceEphemeris::process_spk()`.

6.3.2.4 MAX_PATH_LENGTH

```
constexpr int MAX_PATH_LENGTH = 129 [static], [constexpr]
```

Definition at line 52 of file `spice_ephem.cc`.

Referenced by `jeod::SpiceEphemeris::process_spk()`.

Chapter 7

Namespace Documentation

7.1 jeod Namespace Reference

Namespace jeod.

Data Structures

- class [SpiceEphemeris](#)
The S_define-level class that provides planetary ephemerides.
- class [SpiceEphemOrientation](#)
A [SpiceEphemOrientation](#) minimally extends EphemerisOrientation to include the JEOD and SPICE names and an update method for the target ephemeris reference frame.
- class [SpiceEphemPoint](#)
A [SpiceEphemPoint](#) minimally extends EphemerisPoint, primarily in order to contain the ID of the SPICE object that will be used to update the state of the target ephemeris reference frame.

7.1.1 Detailed Description

Namespace jeod.

Chapter 8

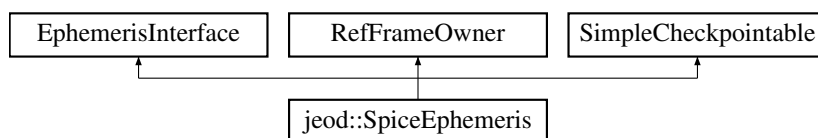
Data Structure Documentation

8.1 jeod::SpiceEphemeris Class Reference

The S_define-level class that provides planetary ephemerides.

```
#include <spice_ephem.hh>
```

Inheritance diagram for jeod::SpiceEphemeris:



Public Member Functions

- `SpiceEphemeris ()`
SpiceEphemeris default constructor.
- `~SpiceEphemeris ()` override
SpiceEphemeris destructor.
- `SpiceEphemeris (const SpiceEphemeris &)=delete`
- `SpiceEphemeris & operator= (const SpiceEphemeris &)=delete`
- `void initialize_model (const TimeManager &time_manager, EphemeridesManager &ephem_manager)`
Initialize the SpiceEphemeris model.
- `void activate ()` override
Nominally, activate the object.
- `void deactivate ()` override
Deactivate the SpiceEphemeris object.
- `double timestamp ()` const override
Return time of last update.
- `std::string get_name ()` const override
Return model name.
- `void ephem_initialize (EphemeridesManager &ephem_manager)` override
Complete the initialization process.
- `void ephem_activate (EphemeridesManager &ephem_manager)` override

- *Mark appropriate items in the model as active.*
- void `ephem_build_tree` (EphemeridesManager &ephem_manager) override
Construct the ephemeris model portions of the reference frame tree.
- void `ephem_update` () override
Update ephemerides for subscribed items.
- void `simple_restore` () override
Set the SPICE model for a restart.
- void `add_planet_name` (std::string planet_name)
- void `add_orientation` (std::string object_name)
- `SpiceEphemPoint * find_spice_id` (int id_to_find)
Find a SPICE ID in the loaded list of SPICE objects.

Data Fields

- std::string `metakernel_filename`
The name of a text file containing the list of SPICE files to be loaded.

Protected Attributes

- bool `inactive` {}
If set to true, makes the model inactive.

Private Member Functions

- `JEOD_CLASS_ESTABLISH_FRIENDS2p` (jeod, `SpiceEphemeris`)
- void `initialize_time` (const TimeManager &time_manager)
Initialize `SpiceEphemeris` timing.
- void `load_spice_files` ()
Load SPICE kernel files containing ephemeris data.
- void `process_spk` ()
Process spk objects and store array of loaded spk IDs.
- void `process_orientations` ()
Load and locate all planetary orientation frames.
- void `introduce_item` (EphemerisItem &item)
Introduce an EphemerisItem to the EphemeridesManager.
- void `populate_item` (EphemerisItem &item, const std::string &name)
Populate basic attributes of a new `SpiceEphemPoint`.
- `SpiceEphemPoint * create_new_ephem_point` (std::string object_name, const std::string &spice_name)
Create a new `SpiceEphemPoint`.
- `SpiceEphemOrientation * create_new_ephem_orientation` (std::string jeod_name)
Create a new `SpiceEphemOrientation`.
- void `initialize_items` ()
Initialize the `SpiceEphemeris` item data.
- std::string `spice_2_jeod` (std::string spice_name)
Convert SPICE names to JEOD nomenclature.
- std::string `jeod_2_spice_pfix` (std::string jeod_name)
Convert JEOD body name to SPICE pfix frame name.
- void `name_barycenter_frames` ()
Name all the planetary barycenters with their JEOD identifiers.

- void [add_barycenter](#) (int spice_id)
Add a barycenter corresponding to the given SPICE ID.
- void [create_barycenters](#) ()
Check what if any barycenters need creating and do so.
- void [determine_root_node](#) ()
Determine which item should be the root of the ref frame tree.
- void [add_descendants_r](#) (SpiceEphemPoint *parent)
Add all offspring from a given node of the ref frame tree.
- int [find_parent_id](#) (int obj_id)
Find the ID of the parent of a given SPICE object.
- void [update_trans](#) ()
Update ephemerides of inertial frames supplied by spk files.
- void [update_rot](#) ()
Update planetary orientations.
- void [mute_spice_errors](#) ()
Mute SPICE errors so that they can be handled by the MessageHandler.

Private Attributes

- bool [force_update](#) {}
Is an update needed even if the time hasn't changed?
- JeodObjectVector< std::string >::type [planet_names](#)
The names of all planets to load into the simulation.
- JeodObjectVector< std::string >::type [orientation_names](#)
The names of all objects for which orientation is required.
- JeodPointerVector< [SpiceEphemPoint](#) >::type [loaded_spk](#)
All the spk objects tracked by this ephemeris.
- JeodPointerVector< [SpiceEphemOrientation](#) >::type [planetary_orientations](#)
Objects defining planet-fixed reference frames.
- std::string [ident](#) {"SPICE"}
Identifier for this model, set by the constructor.
- double [update_time](#) {-99e99}
Time of last update, dynamic time seconds.
- EphemerisRefFrame [barycenter_frames](#) [10]
Array of barycenter frames corresponding to SSBary and the planets.
- [SpiceEphemPoint](#) * [root_item](#) {}
The root point in the reference frame tree.
- const double * [tdb_seconds](#) {}
The source of ephemeris time information.
- const double * [dyn_seconds](#) {}
The source of dynamic time information.
- EphemeridesManager * [ephem_mgr_local](#) {}
Local ephemeris manager pointer to eliminate cascading passes in methods where possible (note some in inherited methods must remain).

8.1.1 Detailed Description

The S_define-level class that provides planetary ephemerides.

The [SpiceEphemeris](#) class constructs the ephemeris reference frame tree and updates the states of the planets based on data from a SPICE file or files.

Definition at line 91 of file `spice_ephem.hh`.

8.1.2 Constructor & Destructor Documentation

8.1.2.1 `SpiceEphemeris()` [1/2]

```
jeod::SpiceEphemeris::SpiceEphemeris ( )
```

[SpiceEphemeris](#) default constructor.

Definition at line 64 of file `spice_ephem.cc`.

References `loaded_spk`, `orientation_names`, `planet_names`, and `planetary_orientations`.

8.1.2.2 `~SpiceEphemeris()`

```
jeod::SpiceEphemeris::~~SpiceEphemeris ( ) [override]
```

[SpiceEphemeris](#) destructor.

Definition at line 76 of file `spice_ephem.cc`.

References `loaded_spk`, `orientation_names`, `planet_names`, and `planetary_orientations`.

8.1.2.3 `SpiceEphemeris()` [2/2]

```
jeod::SpiceEphemeris::SpiceEphemeris (
    const SpiceEphemeris & ) [delete]
```

8.1.3 Member Function Documentation

8.1.3.1 `activate()`

```
void jeod::SpiceEphemeris::activate ( ) [override]
```

Nominally, activate the object.

In the case of a [SpiceEphemeris](#) object, an inactive object cannot be activated once the simulation starts. Note that a [SpiceEphemeris](#) is active by default.

Definition at line 106 of file `spice_ephem.cc`.

References `inactive`.

8.1.3.2 `add_barycenter()`

```
void jeod::SpiceEphemeris::add_barycenter (
    int id ) [private]
```

Add a barycenter corresponding to the given SPICE ID.

Parameters

<i>in</i>	<i>id</i>	id of barycenter to add
-----------	-----------	-------------------------

Definition at line 601 of file `spice_ephem.cc`.

References `barycenter_frames`, `create_new_ephem_point()`, `ephem_mgr_local`, `loaded_spk`, and `MAX_NAME_LENGTH`.

Referenced by `create_barycenters()`.

8.1.3.3 add_descendants_r()

```
void jeod::SpiceEphemeris::add_descendants_r (
    SpiceEphemPoint * parent ) [private]
```

Add all offspring from a given node of the ref frame tree.

Parameters

<i>in, out</i>	<i>parent</i>	Frame to attach to
----------------	---------------	--------------------

Definition at line 756 of file `spice_ephem.cc`.

References `ephem_mgr_local`, `get_name()`, `jeod::SpiceEphemPoint::get_spice_id()`, and `loaded_spk`.

Referenced by `ephem_build_tree()`.

8.1.3.4 add_orientation()

```
void jeod::SpiceEphemeris::add_orientation (
    std::string object_name ) [inline]
```

Definition at line 128 of file `spice_ephem.hh`.

References `orientation_names`.

8.1.3.5 add_planet_name()

```
void jeod::SpiceEphemeris::add_planet_name (
    std::string planet_name ) [inline]
```

Definition at line 121 of file `spice_ephem.hh`.

References `planet_names`.

8.1.3.6 create_barycenters()

```
void jeod::SpiceEphemeris::create_barycenters ( ) [private]
```

Check what if any barycenters need creating and do so.

Any new barycenters are added to the all_loaded_items list and registered with the dynamics manager.

Definition at line 628 of file spice_ephem.cc.

References add_barycenter(), and loaded_spk.

Referenced by initialize_items().

8.1.3.7 create_new_ephem_orientation()

```
SpiceEphemOrientation * jeod::SpiceEphemeris::create_new_ephem_orientation (
    std::string jeod_name ) [private]
```

Create a new [SpiceEphemOrientation](#).

Returns

Pointer to object

Parameters

in	<i>jeod_name</i>	Name of new object
----	------------------	--------------------

Definition at line 468 of file spice_ephem.cc.

References jeod_2_spice_pfix(), populate_item(), jeod::SpiceEphemOrientation::set_spice_frame_name(), tdb_↵ seconds, and jeod::SpiceEphemOrientation::validate().

Referenced by process_orientations().

8.1.3.8 create_new_ephem_point()

```
SpiceEphemPoint * jeod::SpiceEphemeris::create_new_ephem_point (
    std::string object_name,
    const std::string & spice_name ) [private]
```

Create a new [SpiceEphemPoint](#).

Returns

Pointer to object

Parameters

in	<i>object_name</i>	Name of new object
in	<i>spice_name</i>	SPICE lookup name

Definition at line 432 of file `spice_ephem.cc`.

References `find_parent_id()`, `populate_item()`, `jeod::SpiceEphemPoint::set_parent_id()`, and `jeod::SpiceEphemPoint::set_spice_id()`.

Referenced by `add_barycenter()`, and `process_spk()`.

8.1.3.9 deactivate()

```
void jeod::SpiceEphemeris::deactivate ( ) [override]
```

Deactivate the [SpiceEphemeris](#) object.

Definition at line 120 of file `spice_ephem.cc`.

References inactive.

8.1.3.10 determine_root_node()

```
void jeod::SpiceEphemeris::determine_root_node ( ) [private]
```

Determine which item should be the root of the ref frame tree.

Definition at line 673 of file `spice_ephem.cc`.

References `find_spice_id()`, `loaded_spk`, and `root_item`.

Referenced by `ephem_build_tree()`, and `initialize_items()`.

8.1.3.11 ephem_activate()

```
void jeod::SpiceEphemeris::ephem_activate (
    EphemeridesManager & ephem_manager ) [override]
```

Mark appropriate items in the model as active.

Parameters

in, out	<i>ephem_manager</i>	Ephemerides manager
---------	----------------------	---------------------

Definition at line 744 of file spice_ephem.cc.

8.1.3.12 ephemeris_build_tree()

```
void jeod::SpiceEphemeris::ephemeris_build_tree (
    EphemeridesManager & ephem_manager ) [override]
```

Construct the ephemeris model portions of the reference frame tree.

Parameters

<i>in, out</i>	<i>ephem_manager</i>	Ephemerides manager
----------------	----------------------	---------------------

Definition at line 788 of file spice_ephem.cc.

References `add_descendants_r()`, `determine_root_node()`, `inactive`, `loaded_spk`, and `root_item`.

8.1.3.13 ephemeris_initialize()

```
void jeod::SpiceEphemeris::ephemeris_initialize (
    EphemeridesManager & ephem_manager ) [override]
```

Complete the initialization process.

This method should be called after all other ephemeris models have completed their basic initialization and after all planets have registered themselves with the ephemeris manager.

Parameters

<i>in, out</i>	<i>ephem_manager</i>	Ephemerides manager
----------------	----------------------	---------------------

Definition at line 732 of file spice_ephem.cc.

8.1.3.14 ephemeris_update()

```
void jeod::SpiceEphemeris::ephemeris_update ( ) [override]
```

Update ephemerides for subscribed items.

Definition at line 835 of file spice_ephem.cc.

References `dyn_seconds`, `force_update`, `inactive`, `update_rot()`, `update_time`, and `update_trans()`.

8.1.3.15 find_parent_id()

```
int jeod::SpiceEphemeris::find_parent_id (
    int obj_id ) [private]
```

Find the ID of the parent of a given SPICE object.

Here "parent" means both:

1. the frame to which the object is connected in the JEOD reference frame tree (the "parent" frame).
2. the object that will be used as the "observer" in the calls to SPICE to obtain the state of the given object.

Returns

of parent
Units: ID

Parameters

in	<i>obj_id</i>	child ID of which parent is to be found
----	---------------	---

Definition at line 881 of file spice_ephem.cc.

Referenced by create_new_ephem_point().

8.1.3.16 find_spice_id()

```
SpiceEphemPoint * jeod::SpiceEphemeris::find_spice_id (
    int id_to_find )
```

Find a SPICE ID in the loaded list of SPICE objects.

Returns

Pointer to object

Parameters

in	<i>id_to_find</i>	SPICE lookup name
----	-------------------	-------------------

Definition at line 857 of file spice_ephem.cc.

References loaded_spk.

Referenced by determine_root_node().

8.1.3.17 get_name()

```
std::string jeod::SpiceEphemeris::get_name ( ) const [override]
```

Return model name.

Returns

Name

Definition at line 138 of file `spice_ephem.cc`.

References `ident`.

Referenced by `add_descendants_r()`, and `update_trans()`.

8.1.3.18 initialize_items()

```
void jeod::SpiceEphemeris::initialize_items ( ) [private]
```

Initialize the [SpiceEphemeris](#) item data.

Definition at line 482 of file `spice_ephem.cc`.

References `create_barycenters()`, `determine_root_node()`, `introduce_item()`, `loaded_spk`, `planetary_orientations`, and `root_item`.

Referenced by `initialize_model()`.

8.1.3.19 initialize_model()

```
void jeod::SpiceEphemeris::initialize_model (
    const TimeManager & time_manager,
    EphemeridesManager & ephem_manager )
```

Initialize the [SpiceEphemeris](#) model.

This method is called before the planets have been registered with the reference frame manager, so we don't know whether the ephemeris items should be enabled or disabled.

Parameters

in	<i>time_manager</i>	Time manager
in	<i>ephem_manager</i>	Incoming ephemeris manager

Definition at line 151 of file `spice_ephem.cc`.

References `ephem_mgr_local`, `inactive`, `initialize_items()`, `initialize_time()`, `load_spice_files()`, `mute_spice_errors()`, `name_barycenter_frames()`, `process_orientations()`, and `process_spk()`.

8.1.3.20 initialize_time()

```
void jeod::SpiceEphemeris::initialize_time (
    const TimeManager & time_manager ) [private]
```

Initialize [SpiceEphemeris](#) timing.

Parameters

in	<i>time_manager</i>	Time manager
----	---------------------	--------------

Definition at line 200 of file `spice_ephem.cc`.

References `dyn_seconds`, and `tdb_seconds`.

Referenced by `initialize_model()`.

8.1.3.21 introduce_item()

```
void jeod::SpiceEphemeris::introduce_item (
    EphemerisItem & item ) [private]
```

Introduce an `EphemerisItem` to the `EphemeridesManager`.

Parameters

in	<i>item</i>	Item to introduce
----	-------------	-------------------

Definition at line 518 of file `spice_ephem.cc`.

References `ephem_mgr_local`.

Referenced by `initialize_items()`.

8.1.3.22 jeod_2_spice_pfix()

```
std::string jeod::SpiceEphemeris::jeod_2_spice_pfix (
    std::string jeod_name ) [private]
```

Convert JEOD body name to SPICE pfix frame name.

Returns

name of pfix frame that SPICE associates with the given body. Handles high precision frames for Earth and Moon unless IAU is specified by user

Parameters

in	<i>jeod_name</i>	Name of the JEOD body.
----	------------------	------------------------

Definition at line 557 of file `spice_ephem.cc`.

Referenced by `create_new_ephem_orientation()`.

8.1.3.23 JEOD_CLASS_ESTABLISH_FRIENDS2p()

```
jeod::SpiceEphemeris::JEOD_CLASS_ESTABLISH_FRIENDS2p (
    jeod ,
    SpiceEphemeris ) [private]
```

8.1.3.24 load_spice_files()

```
void jeod::SpiceEphemeris::load_spice_files ( ) [private]
```

Load SPICE kernel files containing ephemeris data.

Definition at line 226 of file `spice_ephem.cc`.

References `MAX_MSG_LENGTH`, and `metakernel_filename`.

Referenced by `initialize_model()`.

8.1.3.25 mute_spice_errors()

```
void jeod::SpiceEphemeris::mute_spice_errors ( ) [private]
```

Mute SPICE errors so that they can be handled by the `MessageHandler`.

Definition at line 978 of file `spice_ephem.cc`.

References `MAX_NAME_LENGTH`.

Referenced by `initialize_model()`.

8.1.3.26 name_barycenter_frames()

```
void jeod::SpiceEphemeris::name_barycenter_frames ( ) [private]
```

Name all the planetary barycenters with their JEOD identifiers.

Definition at line 575 of file spice_ephem.cc.

References `barycenter_frames`, `MAX_NAME_LENGTH`, and `spice_2_jeod()`.

Referenced by `initialize_model()`.

8.1.3.27 operator=()

```
SpiceEphemeris& jeod::SpiceEphemeris::operator= (
    const SpiceEphemeris & ) [delete]
```

8.1.3.28 populate_item()

```
void jeod::SpiceEphemeris::populate_item (
    EphemerisItem & item,
    const std::string & object_name ) [private]
```

Populate basic attributes of a new [SpiceEphemPoint](#).

Parameters

<code>in, out</code>	<i>item</i>	Pointer to item to populate
<code>in</code>	<i>object_name</i>	Name of the item

Definition at line 419 of file spice_ephem.cc.

Referenced by `create_new_ephem_orientation()`, and `create_new_ephem_point()`.

8.1.3.29 process_orientations()

```
void jeod::SpiceEphemeris::process_orientations ( ) [private]
```

Load and locate all planetary orientation frames.

Definition at line 401 of file spice_ephem.cc.

References `create_new_ephem_orientation()`, `orientation_names`, and `planetary_orientations`.

Referenced by `initialize_model()`.

8.1.3.30 process_spk()

```
void jeod::SpiceEphemeris::process_spk ( ) [private]
```

Process spk objects and store array of loaded spk IDs.

Definition at line 252 of file spice_ephem.cc.

References `create_new_ephem_point()`, `loaded_spk`, `MAX_IDS`, `MAX_NAME_LENGTH`, `MAX_PATH_LENGTH`, and `planet_names`.

Referenced by `initialize_model()`.

8.1.3.31 simple_restore()

```
void jeod::SpiceEphemeris::simple_restore ( ) [override]
```

Set the SPICE model for a restart.

Definition at line 191 of file spice_ephem.cc.

8.1.3.32 spice_2_jeod()

```
std::string jeod::SpiceEphemeris::spice_2_jeod (
    std::string spice_name ) [private]
```

Convert SPICE names to JEOD nomenclature.

Returns

JEOD-friendly name of a SPICE name

Parameters

in	<i>spice_name</i>	Name of a SPICE object.
----	-------------------	-------------------------

Definition at line 542 of file spice_ephem.cc.

Referenced by `name_barycenter_frames()`.

8.1.3.33 timestamp()

```
double jeod::SpiceEphemeris::timestamp ( ) const [override]
```

Return time of last update.

Returns

Timestamp
Units: day

Definition at line 129 of file spice_ephem.cc.

References `update_time`.

8.1.3.34 update_rot()

```
void jeod::SpiceEphemeris::update_rot ( ) [private]
```

Update planetary orientations.

Definition at line 967 of file spice_ephem.cc.

References `dyn_seconds`, `planetary_orientations`, and `tdb_seconds`.

Referenced by `ephem_update()`.

8.1.3.35 update_trans()

```
void jeod::SpiceEphemeris::update_trans ( ) [private]
```

Update ephemerides of inertial frames supplied by spk files.

Definition at line 906 of file spice_ephem.cc.

References `get_name()`, `loaded_spk`, `MAX_MSG_LENGTH`, `root_item`, `tdb_seconds`, and `update_time`.

Referenced by `ephem_update()`.

8.1.4 Field Documentation

8.1.4.1 barycenter_frames

```
EphemerisRefFrame jeod::SpiceEphemeris::barycenter_frames[10] [private]
```

Array of barycenter frames corresponding to SSBary and the planets.

`trick_units(-)`

Definition at line 197 of file spice_ephem.hh.

Referenced by `add_barycenter()`, and `name_barycenter_frames()`.

8.1.4.2 dyn_seconds

```
const double* jeod::SpiceEphemeris::dyn_seconds {} [private]
```

The source of dynamic time information.

trick_units(-)

Definition at line 212 of file spice_ephem.hh.

Referenced by ephem_update(), initialize_time(), and update_rot().

8.1.4.3 ephem_mgr_local

```
EphemeridesManager* jeod::SpiceEphemeris::ephem_mgr_local {} [private]
```

Local ephem manager pointer to eliminate cascading passes in methods where possible (note some in inherited methods must remain).

trick_units(-)

Definition at line 218 of file spice_ephem.hh.

Referenced by add_barycenter(), add_descendants_r(), initialize_model(), and introduce_item().

8.1.4.4 force_update

```
bool jeod::SpiceEphemeris::force_update {} [private]
```

Is an update needed even if the time hasn't changed?

trick_units(-)

Definition at line 162 of file spice_ephem.hh.

Referenced by ephem_update().

8.1.4.5 ident

```
std::string jeod::SpiceEphemeris::ident {"SPICE"} [private]
```

Identifier for this model, set by the constructor.

trick_units(-)

Definition at line 187 of file spice_ephem.hh.

Referenced by get_name().

8.1.4.6 inactive

```
bool jeod::SpiceEphemeris::inactive {} [protected]
```

If set to true, makes the model inactive.

This is set to false (i.e., model is active) by the constructor. Setting this flag to true prior to initialization time will result in the model never doing anything. Setting this flag to true after the model has been active for some time turns it off from that point forward. Turning on the model after it has been inactive for some time is not supported.

Definition at line 154 of file `spice_ephem.hh`.

Referenced by `activate()`, `deactivate()`, `ephem_build_tree()`, `ephem_update()`, and `initialize_model()`.

8.1.4.7 loaded_spk

```
JeodPointerVector<SpiceEphemPoint>::type jeod::SpiceEphemeris::loaded_spk [private]
```

All the spk objects tracked by this ephemeris.

```
trick_io(**)
```

Definition at line 177 of file `spice_ephem.hh`.

Referenced by `add_barycenter()`, `add_descendants_r()`, `create_barycenters()`, `determine_root_node()`, `ephem_build_tree()`, `find_spice_id()`, `initialize_items()`, `process_spk()`, `SpiceEphemeris()`, `update_trans()`, and `~SpiceEphemeris()`.

8.1.4.8 metakernel_filename

```
std::string jeod::SpiceEphemeris::metakernel_filename
```

The name of a text file containing the list of SPICE files to be loaded.

Must be in format needed by `SPICE furnsh_c()` function. See comments for method `load_spice_files()` in `spice_ephem.cc` for format details. `trick_units(-)`

Definition at line 143 of file `spice_ephem.hh`.

Referenced by `load_spice_files()`.

8.1.4.9 orientation_names

```
JeodObjectVector<std::string>::type jeod::SpiceEphemeris::orientation_names [private]
```

The names of all objects for which orientation is required.

trick_io(**)

Definition at line 172 of file spice_ephem.hh.

Referenced by add_orientation(), process_orientations(), SpiceEphemeris(), and ~SpiceEphemeris().

8.1.4.10 planet_names

```
JeodObjectVector<std::string>::type jeod::SpiceEphemeris::planet_names [private]
```

The names of all planets to load into the simulation.

trick_io(**)

Definition at line 167 of file spice_ephem.hh.

Referenced by add_planet_name(), process_spk(), SpiceEphemeris(), and ~SpiceEphemeris().

8.1.4.11 planetary_orientations

```
JeodPointerVector<SpiceEphemOrientation>::type jeod::SpiceEphemeris::planetary_orientations  
[private]
```

Objects defining planet-fixed reference frames.

trick_io(**)

Definition at line 182 of file spice_ephem.hh.

Referenced by initialize_items(), process_orientations(), SpiceEphemeris(), update_rot(), and ~SpiceEphemeris().

8.1.4.12 root_item

```
SpiceEphemPoint* jeod::SpiceEphemeris::root_item {} [private]
```

The root point in the reference frame tree.

trick_units(-)

Definition at line 202 of file spice_ephem.hh.

Referenced by determine_root_node(), ephem_build_tree(), initialize_items(), and update_trans().

8.1.4.13 tdb_seconds

```
const double* jeod::SpiceEphemeris::tdb_seconds {} [private]
```

The source of ephemeris time information.

trick_units(-)

Definition at line 207 of file spice_ephem.hh.

Referenced by create_new_ephem_orientation(), initialize_time(), update_rot(), and update_trans().

8.1.4.14 update_time

```
double jeod::SpiceEphemeris::update_time {-99e99} [private]
```

Time of last update, dynamic time seconds.

trick_units(s)

Definition at line 192 of file spice_ephem.hh.

Referenced by ephem_update(), timestamp(), and update_trans().

The documentation for this class was generated from the following files:

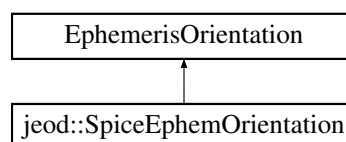
- [spice_ephem.hh](#)
- [spice_ephem.cc](#)

8.2 jeod::SpiceEphemOrientation Class Reference

A [SpiceEphemOrientation](#) minimally extends EphemerisOrientation to include the JEOD and SPICE names and an update method for the target ephemeris reference frame.

```
#include <spice_ephem_orient.hh>
```

Inheritance diagram for jeod::SpiceEphemOrientation:



Public Member Functions

- [SpiceEphemOrientation](#) ()
SpiceEphemOrientation default constructor.
- [~SpiceEphemOrientation](#) () override=default
- [SpiceEphemOrientation](#) (const [SpiceEphemOrientation](#) &)=delete
- [SpiceEphemOrientation](#) & [operator=](#) (const [SpiceEphemOrientation](#) &)=delete
- void [update](#) (double time_tdb, double time_dyn)
Update the rotational state of the target frame.
- void [validate](#) (double time_tdb)
Confirm that the target frame exists in the loaded SPICE kernels.
- void [get_spice_transformation](#) (double time_tdb, double trans6x6[6][6])
Populate the SPICE 6 x 6 matrix via sxform_c().
- void [set_spice_frame_name](#) (const std::string &new_name)
Setter for the name of the SPICE frame.

Private Member Functions

- JEOD_CLASS_ESTABLISH_FRIENDS2 p (jeod, [SpiceEphemOrientation](#))

Private Attributes

- std::string [spice_frame_name](#)
SPICE name of the target reference frame.

8.2.1 Detailed Description

A [SpiceEphemOrientation](#) minimally extends EphemerisOrientation to include the JEOD and SPICE names and an update method for the target ephemeris reference frame.

Definition at line 83 of file spice_ephem_orient.hh.

8.2.2 Constructor & Destructor Documentation

8.2.2.1 [SpiceEphemOrientation](#)() [1/2]

```
jeod::SpiceEphemOrientation::SpiceEphemOrientation ( )
```

[SpiceEphemOrientation](#) default constructor.

Definition at line 44 of file spice_ephem_orient.cc.

8.2.2.2 ~SpiceEphemOrientation()

```
jeod::SpiceEphemOrientation::~~SpiceEphemOrientation ( ) [override], [default]
```

8.2.2.3 SpiceEphemOrientation() [2/2]

```
jeod::SpiceEphemOrientation::SpiceEphemOrientation (
    const SpiceEphemOrientation & ) [delete]
```

8.2.3 Member Function Documentation

8.2.3.1 get_spice_transformation()

```
void jeod::SpiceEphemOrientation::get_spice_transformation (
    double time_tdb,
    double trans6x6[6][6] )
```

Populate the SPICE 6 x 6 matrix via `sxform_c()`.

Parameters

in	<i>time_tdb</i>	Ephem time (TDB)
out	<i>trans6x6</i>	Spice matrix

Definition at line 115 of file `spice_ephem_orient.cc`.

References `spice_frame_name`.

Referenced by `update()`, and `validate()`.

8.2.3.2 operator=()

```
SpiceEphemOrientation& jeod::SpiceEphemOrientation::operator= (
    const SpiceEphemOrientation & ) [delete]
```

8.2.3.3 p()

```
JEOD_CLASS_ESTABLISH_FRIENDS2 jeod::SpiceEphemOrientation::p (
    jeod ,
    SpiceEphemOrientation ) [private]
```

8.2.3.4 set_spice_frame_name()

```
void jeod::SpiceEphemOrientation::set_spice_frame_name (
    const std::string & new_name ) [inline]
```

Setter for the name of the SPICE frame.

Parameters

<i>new_name</i>	Name of the SPICE frame
-----------------	-------------------------

Definition at line 105 of file spice_ephem_orient.hh.

Referenced by jeod::SpiceEphemeris::create_new_ephem_orientation().

8.2.3.5 update()

```
void jeod::SpiceEphemOrientation::update (
    double time_tdb,
    double time_dyn )
```

Update the rotational state of the target frame.

Parameters

in	<i>time_tdb</i>	Ephemeris time (TDB) Units: s
in	<i>time_dyn</i>	dyn time for timestamp Units: s

Definition at line 54 of file spice_ephem_orient.cc.

References get_spice_transformation().

8.2.3.6 validate()

```
void jeod::SpiceEphemOrientation::validate (
    double time_tdb )
```

Confirm that the target frame exists in the loaded SPICE kernels.

Parameters

in	<i>time_tdb</i>	Ephemeris time (TDB)
----	-----------------	----------------------

Definition at line 99 of file spice_ephem_orient.cc.

References `get_spice_transformation()`.

Referenced by `jeod::SpiceEphemeris::create_new_ephem_orientation()`.

8.2.4 Field Documentation

8.2.4.1 spice_frame_name

```
std::string jeod::SpiceEphemOrientation::spice_frame_name [private]
```

SPICE name of the target reference frame.

`trick_units(-)`

Definition at line 115 of file `spice_ephem_orient.hh`.

Referenced by `get_spice_transformation()`.

The documentation for this class was generated from the following files:

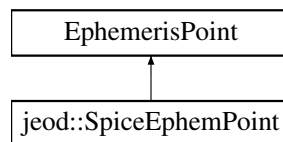
- [spice_ephem_orient.hh](#)
- [spice_ephem_orient.cc](#)

8.3 jeod::SpiceEphemPoint Class Reference

A [SpiceEphemPoint](#) minimally extends `EphemerisPoint`, primarily in order to contain the ID of the SPICE object that will be used to update the state of the target ephemeris reference frame.

```
#include <spice_ephem_point.hh>
```

Inheritance diagram for `jeod::SpiceEphemPoint`:



Public Types

- enum [Status](#) { [IsRoot](#) = 0 , [InTree](#) = 1 , [Active](#) = 2 }
- Enumerates the status values of a [SpiceEphemPoint](#).*

Public Member Functions

- [SpiceEphemPoint](#) ()
[SpiceEphemPoint](#) default constructor.
- [~SpiceEphemPoint](#) () override=default
- [SpiceEphemPoint](#) (const [SpiceEphemPoint](#) &)=delete
- [SpiceEphemPoint](#) & operator= (const [SpiceEphemPoint](#) &)=delete
- virtual void [set_status](#) ([SpiceEphemPoint::Status](#) new_status)
Set the active status.
- virtual [SpiceEphemPoint::Status](#) [get_status](#) () const
Return current status.
- virtual void [set_spice_id](#) (int new_id)
Set ID of associated SPICE kernel object.
- virtual int [get_spice_id](#) () const
Return ID of associated SPICE kernel object.
- virtual void [set_parent_id](#) (int new_id)
Set ID of associated parent SPICE kernel object.
- virtual int [get_parent_id](#) () const
Return ID of associated parent SPICE kernel object.

Protected Attributes

- [Status](#) status {[Active](#)}
The status for the ephemeris reference frame associated with this item.
- int [spice_id](#) {32767}
The SPICE kernel object to be used to maintain the target frame's state.
- int [parent_id](#) {32767}
The SPICE ID of the parent to this object.

Private Member Functions

- JEOD_CLASS_ESTABLISH_FRIENDS2 p (jeod, [SpiceEphemPoint](#))

8.3.1 Detailed Description

A [SpiceEphemPoint](#) minimally extends [EphemerisPoint](#), primarily in order to contain the ID of the SPICE object that will be used to update the state of the target ephemeris reference frame.

Definition at line 82 of file [spice_ephem_point.hh](#).

8.3.2 Member Enumeration Documentation

8.3.2.1 Status

```
enum jeod::SpiceEphemPoint::Status
```

Enumerates the status values of a [SpiceEphemPoint](#).

Enumerator

IsRoot	The target reference frame is extant, active, and is the root of the reference frame tree. Hence its state is the trivial state.
InTree	The target reference frame is extant and active. However, present responsibility for updating the frame lies with some other ephemeris model.
Active	The target reference frame is extant, active, and is to be updated by this ephemeris model.

Definition at line 90 of file `spice_ephem_point.hh`.

8.3.3 Constructor & Destructor Documentation

8.3.3.1 SpiceEphemPoint() [1/2]

```
jeod::SpiceEphemPoint::SpiceEphemPoint ( )
```

[SpiceEphemPoint](#) default constructor.

Definition at line 40 of file `spice_ephem_point.cc`.

8.3.3.2 ~SpiceEphemPoint()

```
jeod::SpiceEphemPoint::~~SpiceEphemPoint ( ) [override], [default]
```

8.3.3.3 SpiceEphemPoint() [2/2]

```
jeod::SpiceEphemPoint::SpiceEphemPoint (
    const SpiceEphemPoint & ) [delete]
```

8.3.4 Member Function Documentation

8.3.4.1 get_parent_id()

```
int jeod::SpiceEphemPoint::get_parent_id ( ) const [virtual]
```

Return ID of associated parent SPICE kernel object.

Returns

Name

Definition at line 94 of file spice_ephem_point.cc.

References parent_id.

8.3.4.2 get_spice_id()

```
int jeod::SpiceEphemPoint::get_spice_id ( ) const [virtual]
```

Return ID of associated SPICE kernel object.

Returns

Name

Definition at line 76 of file spice_ephem_point.cc.

References spice_id.

Referenced by jeod::SpiceEphemeris::add_descendants_r().

8.3.4.3 get_status()

```
SpiceEphemPoint::Status jeod::SpiceEphemPoint::get_status ( ) const [virtual]
```

Return current status.

Returns

Current object status

Definition at line 58 of file spice_ephem_point.cc.

References status.

8.3.4.4 operator=()

```
SpiceEphemPoint& jeod::SpiceEphemPoint::operator= (
    const SpiceEphemPoint & ) [delete]
```

8.3.4.5 p()

```
JEOD_CLASS_ESTABLISH_FRIENDS2 jeod::SpiceEphemPoint::p (
    jeod ,
    SpiceEphemPoint ) [private]
```

8.3.4.6 set_parent_id()

```
void jeod::SpiceEphemPoint::set_parent_id (
    int new_id ) [virtual]
```

Set ID of associated parent SPICE kernel object.

Parameters

in	<i>new</i> ↔ _id	New SPICE ID
----	---------------------	--------------

Definition at line 85 of file spice_ephem_point.cc.

References parent_id.

Referenced by jeod::SpiceEphemeris::create_new_ephem_point().

8.3.4.7 set_spice_id()

```
void jeod::SpiceEphemPoint::set_spice_id (
    int new_id ) [virtual]
```

Set ID of associated SPICE kernel object.

Parameters

in	<i>new</i> ↔ _id	New SPICE ID
----	---------------------	--------------

Definition at line 67 of file spice_ephem_point.cc.

References `spice_id`.

Referenced by `jeod::SpiceEphemeris::create_new_ephem_point()`.

8.3.4.8 `set_status()`

```
void jeod::SpiceEphemPoint::set_status (
    SpiceEphemPoint::Status new_status ) [virtual]
```

Set the active status.

Parameters

in	<i>new_status</i>	New status value
----	-------------------	------------------

Definition at line 49 of file `spice_ephem_point.cc`.

References `status`.

8.3.5 Field Documentation

8.3.5.1 `parent_id`

```
int jeod::SpiceEphemPoint::parent_id {32767} [protected]
```

The SPICE ID of the parent to this object.

`trick_units(-)`

Definition at line 134 of file `spice_ephem_point.hh`.

Referenced by `get_parent_id()`, and `set_parent_id()`.

8.3.5.2 `spice_id`

```
int jeod::SpiceEphemPoint::spice_id {32767} [protected]
```

The SPICE kernel object to be used to maintain the target frame's state.

`trick_units(-)`

Definition at line 129 of file `spice_ephem_point.hh`.

Referenced by `get_spice_id()`, and `set_spice_id()`.

8.3.5.3 status

`Status jeod::SpiceEphemPoint::status {Active} [protected]`

The status for the ephemeris reference frame associated with this item.

`trick_units(-)`

Definition at line 124 of file `spice_ephem_point.hh`.

Referenced by `get_status()`, and `set_status()`.

The documentation for this class was generated from the following files:

- [spice_ephem_point.hh](#)
- [spice_ephem_point.cc](#)

Chapter 9

File Documentation

9.1 spice_ephem.cc File Reference

Define the methods for the SPICE ephemeris model class.

```
#include <algorithm>
#include <cstdint>
#include <fstream>
#include <set>
#include "SpiceUsr.h"
#include "environment/ephemerides/ephem_interface/include/ephem_messages.h"
#include "environment/time/include/time_manager.hh"
#include "utils/memory/include/jeod_alloc.hh"
#include "utils/message/include/message_handler.hh"
#include "utils/named_item/include/named_item.hh"
#include "../include/spice_ephem.hh"
```

Namespaces

- [jeod](#)
Namespace jeod.

Variables

- static constexpr int [MAX_PATH_LENGTH](#) = 129
- static constexpr int [MAX_NAME_LENGTH](#) = 33
- static constexpr int [MAX_MSG_LENGTH](#) = 1841
- static constexpr int [MAX_IDS](#) = 1000

9.1.1 Detailed Description

Define the methods for the SPICE ephemeris model class.

9.2 spice_ephem.hh File Reference

Define class for the SPICE ephemeris model.

```
#include <string>
#include "environment/ephemerides/ephem_interface/include/ephem_interface.↵
hh"
#include "environment/ephemerides/ephem_interface/include/ephem_ref_frame.↵
hh"
#include "environment/ephemerides/ephem_manager/include/ephem_manager.hh"
#include "environment/time/include/class_declarations.hh"
#include "utils/container/include/object_vector.hh"
#include "utils/container/include/pointer_vector.hh"
#include "utils/container/include/simple_checkpointable.hh"
#include "utils/ref_frames/include/ref_frame_interface.hh"
#include "utils/sim_interface/include/jeod_class.hh"
#include "spice_ephem_orient.hh"
#include "spice_ephem_point.hh"
```

Data Structures

- class [jeod::SpiceEphemeris](#)
The S_define-level class that provides planetary ephemerides.

Namespaces

- [jeod](#)
Namespace jeod.

9.2.1 Detailed Description

Define class for the SPICE ephemeris model.

9.3 spice_ephem_orient.cc File Reference

Define the methods for the SPICE-specific ephemeris orientation class.

```
#include "SpiceUsr.h"
#include "environment/ephemerides/ephem_interface/include/ephem_messages.↵
hh"
#include "utils/math/include/matrix3x3.hh"
#include "utils/math/include/vector3.hh"
#include "../include/spice_ephem_orient.hh"
```

Namespaces

- [jeod](#)
Namespace jeod.

9.3.1 Detailed Description

Define the methods for the SPICE-specific ephemeris orientation class.

9.4 spice_ephem_orient.hh File Reference

Define class SpiceEphemOrient, which extends EphemerisOrientation for use with the SPICE ephemeris model.

```
#include <string>
#include "environment/ephemerides/ephem_item/include/ephem_orient.hh"
#include "utils/sim_interface/include/jeod_class.hh"
```

Data Structures

- class [jeod::SpiceEphemOrientation](#)

A [SpiceEphemOrientation](#) minimally extends *EphemerisOrientation* to include the *JEOD* and *SPICE* names and an update method for the target ephemeris reference frame.

Namespaces

- [jeod](#)

Namespace *jeod*.

9.4.1 Detailed Description

Define class SpiceEphemOrient, which extends EphemerisOrientation for use with the SPICE ephemeris model.

9.5 spice_ephem_point.cc File Reference

Define the methods for the SPICE-specific ephemeris point class.

```
#include <string>
#include "utils/memory/include/jeod_alloc.hh"
#include "../include/spice_ephem_point.hh"
```

Namespaces

- [jeod](#)

Namespace *jeod*.

9.5.1 Detailed Description

Define the methods for the SPICE-specific ephemeris point class.

9.6 spice_ephem_point.hh File Reference

Define class SpiceEphemPoint, which extends EphemerisPoint for use with the SPICE ephemeris model.

```
#include "environment/ephemerides/ephem_item/include/ephem_point.hh"  
#include "utils/sim_interface/include/jeod_class.hh"
```

Data Structures

- class [jeod::SpiceEphemPoint](#)

A [SpiceEphemPoint](#) minimally extends *EphemerisPoint*, primarily in order to contain the ID of the SPICE object that will be used to update the state of the target ephemeris reference frame.

Namespaces

- [jeod](#)

Namespace *jeod*.

9.6.1 Detailed Description

Define class SpiceEphemPoint, which extends EphemerisPoint for use with the SPICE ephemeris model.

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